as [Noble gas]ns^2 where 'n' represents the valence shell.

Table 5.8 Electronic configuration of alkaline earth metals

Element	Atomic No.	Electronic configuration
Be	4	$[He]2s^2$
Mg	12	$[Ne]3s^2$
Ca	20	$[Ar]4s^2$
Sr	38	$[Kr]5s^2$
Ba	56	$[Xe]6s^2$
Ra	88	$[Rn]7s^2$

Atomic and ionic radii

The atomic and ionic radii of alkaline earth metals are smaller than the corresponding members of the alkali metals. This is due to the fact the Group 2 elements having a higher nuclear charge that allows electrons to be attracted more strongly towards the nucleus. On moving down the group, the radii increases due to gradual increase in the number of the shells and the screening effect.

Common oxidation state

The group 2 elements have two electrons in their valence shell and by losing these electrons, they acquire the stable noble gas configuration. So these elements exhibit +2 oxidation state in their compounds.

Ionisation enthalpy

Due to a fairly large size of the atoms, alkaline earth metals have low ionisation enthalpies when compared to 'p' block elements. Down the group the ionisation enthalpy decreases as atomic size increases. This is due to the addition of new shells as well as increase in the magnitude of the screening effect of inner shell electrons. Members of group 2 have higher ionization enthalpy values than group 1 because of their smaller size, with electrons being more attracted towards the nucleus of the atoms. Correspondingly they are less electropositive than alkali metals.

Table 5.9 Physical properties of alkaline earth metals

Physical property	Be	Mg	Ca	Sr	Ba
Atomic radius -non bonded (Å)	1.12	1.60	1.97	2.15	2.22
Ionic radius (Å)	0.27(0.31)	0.72	1.00	1.18	1.35
First ionization energy (kJ mol^{-1})	899.5	737.8	589.8	549.5	502.9
Second ionization energy (kJ mol^{-1})	1757.1	1450.7	1145.5	1064.2	965.2
Hydration enthalpy (kJ mol^{-1})	– 2494	– 1921	–1577	– 1443	– 1305
Melting Point (^{0} C)	1287	651	851	789	729
Boiling Point (^{0} C)	2472	1090	1494	1382	1805
Density (g cm^{-3})	1.84	1.74	1.55	2.63	3.59
Standard Potential $E^{\ominus}$ for M^+/M (V)	-1.97	-2.36	-2.84	-2.89	-2.92
Electronegativity (Paulings scale)	1.6	1.2	1.0	1.0	0.9

11th Std Chemistry 125-157.indd 137 06-01-2021 15:29:15